A Certified Finite-Cutoff Program for the SU(2) Theta-Graph Transfer Operator: Interacting Positivity, Exact Seam-Integer Counting, and a First Uniformity-in-the-Cutoff Theorem

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We report a finite, machine-certified program studying the spectral gap of the interacting transfer operator on the smallest non-trivial closed graph with a plaquette — the theta graph, two vertices joined by three edges — for SU(2) lattice gauge theory. Every numerical statement below is an exact two-sided rational (or exact-integer) enclosure produced by directed interval arithmetic or exact linear algebra, never a floating-point estimate; every claim carries a machine-checked certificate reproducible from the accompanying repository. We prove: (i) the free reduced gap in closed form via the exact SU(2) character spectrum; (ii) a certified two-sided interacting gap at every coupling $\kappa \geq 0$ on this graph, via a Kill-Lemma-shaped decomposition into a block Schur estimate and a positivity statement, both certified rather than assumed; (iii) that the first-order crossing direction is an exact rank-one, graph-symmetry –protected line, not an artifact of the compression used to compute it; (iv) an exact eigenvalue-counting reformulation of the gap via a Haynsworth-congruence (Birman–Schwinger-type) construction, giving exact integer spectral counts rather than numerical brackets on part of the coupling range; (v) the program's first statement that is uniform in the lattice cutoff rather than proved at one fixed cutoff, obtained by replacing the Wilson action with the heat-kernel action, whose exact semigroup property removes discretization error identically; and (vi) that the interacting gap is volume- and cutoff-uniform simultaneously on a family of graphs built by vertex-gluing theta graphs with no shared plaquettes, while the same route provably fails once plaquettes are shared, localizing the residual volume obstruction exactly to interaction overlap. We are explicit about what this program does not establish: it does not touch the infinite-volume, continuum, or Osterwalder–Schrader-reconstruction content of the Yang–Mills existence-and-mass-gap problem, and none of the results here should be read as bearing on that problem beyond providing an exactly verified finite testbed and a catalogue of which certification routes do and do not survive volume growth.

I. INTRODUCTION

The Yang–Mills existence and mass-gap problem asks for a proof that a non-trivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ satisfying the Wightman/Osterwalder–Schrader axioms, with a mass gap $\Delta > 0$ [1]. That statement concerns the continuum, infinite-volume limit of the theory. The present paper does not attempt it, and no theorem below should be misread as a step toward the Clay-sense predicate.

What we do attempt is narrower and, we argue, still useful: to take the smallest closed graph carrying a non-trivial plaquette — the theta graph B_3, two vertices joined by three edges, one independent holonomy after gauge fixing, three faces — and certify, with exact arithmetic and machine-checked proofs, every spectral statement that can honestly be made about its interacting transfer operator at finite lattice spacing. The discipline we impose throughout is: *no floating-point number is ever part of a verdict.* Every enclosure below is either an exact rational computation or a two-sided interval of provably bounded width obtained from directed rounding, and every theorem is accompanied by an executable certificate, a set of negative controls that must fail when the underlying claim is deliberately falsified, and an explicit statement of what remains open. This is closer in spirit to interval-certified numerics and to formally verified mathematics than to a physics estimate, and we adopt it because the target problem is famous partly for how easy it is to overclaim.

The paper is organized as a linear certification chain, YM-1 through YM-14, each step consuming only what the previous steps proved. We give theorem statements, proof sketches sufficient for a reader to reconstruct the argument, and pointers to the exact certificate in the accompanying repository for full reproducibility. Section III fixes the graph, the gauge-fixed carrier, and the character-expansion toolkit. Section IV treats the free (non-interacting) reduced gap. Section V builds the interacting gap at finite coupling on the theta graph in stages: a sandwich bound, the first-order crossing direction, a symmetry-protection theorem, a certified two-sided gap beyond the sandwich's reach, and an exact eigenvalue-counting reformulation. Section VI states the capstone theorem: a strictly positive gap at every coupling on the fixed graph, together with the honest statement of what "every coupling, fixed graph" does not give. Section VII gives the paper's central positive result on the harder axis: a gap that is uniform in the lattice cutoff. Section VIII extends this along the volume axis on a restricted family and localizes exactly where the extension

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fails. Section IX audits three structural gates — gauge invariance, volume/infrared behaviour, and regulator universality — and reports two of the three as genuinely split between a closed and an open half. Section XI states the remaining obligations without euphemism. Section XII concludes.

II. METHODOLOGY: CERTIFICATION AND REPRODUCIBILITY

Because this program's value depends entirely on a reader's ability to check it independently rather than take it on trust, we state the certification methodology precisely before using it.

Exact and directed-interval arithmetic only. No floating-point number ever appears inside a verdict. Rational quantities (eigenvalue multiplicities, Gram-matrix overlaps, seam-integer counts) are computed with Python's arbitrary-precision `fractions.Fraction` and carried exactly; no external numerical library is required for these. Transcendental quantities (Bessel functions I_ν, logarithms) are bracketed by two-sided rational intervals $[\ell, u]$ with $\ell \leq x \leq u$ guaranteed by construction: I_ν is enclosed by its everywhere-positive power series truncated at a declared order, with a geometric-series majorant bounding the truncated tail from above; log is enclosed via the atanh series, which is exact-sign alternation-free and admits the same majorant technique. Every arithmetic operation on such intervals rounds outward (down for lower bounds, up for upper bounds), so the true value is guaranteed to remain inside the bracket at every step, and every reported enclosure states its width explicitly.

Hash-pinned certificates. Each theorem above has one corresponding executable script (Table I) that recomputes its result from these primitives and writes a certificate file containing the exact value or bracket together with a SHA-256 digest of the full certificate content. That digest is checked against an `EXPECTED_*.sha256` file committed alongside it. A change to any input, any arithmetic step, or any stated conclusion changes the digest and is caught automatically; nothing about a certified result can be edited by hand without the mismatch being visible.

Negative controls. Every script includes at least one deliberately falsified variant of its own computation — a wrong Bessel recurrence, a substituted coupling, a sign flip on the disputed term, a shrunk carrier known to be too small — and asserts that this variant produces a *different, separated* bracket or a different exact count from the certified one. A script fails if a negative control does not separate, which is the mechanism that distinguishes a theorem that constrains the actual object from an identity that would hold on arbitrary data (the discipline we call, in the repository, the *renaming test*: before a computed coincidence is reported as a finding about the carrier, we check that it disappears under repetition on random data of the same shape).

TABLE I. Theorem-to-script correspondence. Every result in this paper is the printed output of the named script, with certificate digest pinned in the correspondingly named `EXPECTED_*.sha256` file.

Theorem	Script (in `certificates/`)
YM-1 (1)	`ym1_certified_gap.py`
YM-2 (2)	`ym2_theta_interacting_gap.py`
YM-3 (3)	`ym3_crossing_direction.py`
YM-4 (4)	`ym4_symmetry_protected.py`
YM-5 (5)	`ym5_two_sided_gap.py`
YM-6/7 (6)	`ym6_seam_integer_dock.py`, `ym7_v7_crossing_curves.py`
YM-8 (7)	`ym8_all_coupling_capstone.py`
YM-9 (8)	`ym9_uniform_heat_kernel.py`
YM-10 (13)	`ym10_blindness_ledger.py`
YM-11 (10–12)	`ym11_gate_verdicts.py`
YM-12 (9)	`ym12_square_sourced.py`
YM-14 (14)	`ym14_overlap_dock.py`

Independent re-execution and continuous integration. Several results above (the Theorem 6 counting law, the Theorem 8 uniformity theorem) were independently re-derived or corrected after a first pass flagged an inconsistency between a narrated claim and the arithmetic the script actually performed; we report such corrections in place (Section IX) rather than silently overwrite the earlier claim. The repository's continuous integration workflow re-executes every certificate script and every test on each change and fails the build on any pin mismatch, so the checked-in state of the repository is, at every commit, the actual output of the actual scripts, not a hand-maintained summary of them.

Reproducing a specific theorem. Any of the results above can be independently checked with no dependency beyond a standard Python 3 interpreter (no NumPy, no computer-algebra system) by cloning the public repository and running the corresponding script, e.g.:

```
git clone https://github.com/Parveen117/
    Publications
cd Publications/papers/
    yang-mills-certified-benchmark
python certificates/ym1_certified_gap.py
python -m pytest tests/ -v
```

Each script prints its certified value or bracket, its digest, and the result of every negative control; the test suite additionally checks every digest against the committed `EXPECTED_*.sha256` file.

III. GRAPH, CARRIER, AND CHARACTER TOOLKIT

Let B_3 be the theta graph: two vertices v_0, v_1 joined by three edges e_A, e_B, e_C with SU(2)-valued holonomies

A, B, C. Tree gauge-fixing on any connected graph is exact and free of Gribov ambiguity: fixing a spanning tree (here, edge e_C, so $C = ⊮$) leaves exactly $b_1(B_3) = E - V + 1 = 2$ independent holonomies, here (A, B), with residual gauge acting diagonally by conjugation. Gauge-invariant states are therefore exactly $L^2(\mathrm{SU}(2)\times\mathrm{SU}(2))^{\mathrm{Ad}}$, and this fact is itself certified rather than assumed (Section IX); every capsule in this paper uses it as the carrier.

We work in the character basis. For $g \in \mathrm{SU}(2)$, the irreducible characters $\chi_j(g)$, $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, satisfy the exact orthogonality and Clebsch–Gordan pairing relations of SU(2); in particular the triple integral

$$\int_{\mathrm{SU}(2)} \chi_p(A)\, \chi_q(B)\, \chi_r(AB^{-1})\, dA\, dB = \frac{\delta_{p=q=r}}{d_p}, \qquad (1)$$

with $d_p = 2p + 1$, is exact and rational, and is the single identity that governs both the intra- and inter-plaquette seam structure used throughout. The Wilson single-link heat kernel has exact character expansion $K_\beta(g) = \sum_j d_j\, \lambda_j(\beta)\, \chi_j(g)$ with $\lambda_j(\beta) = I_{2j+1}(\beta)/I_1(\beta)$ (modified Bessel functions), and the true heat-kernel (semigroup) action used from Section VII on has exact expansion coefficients $f_j(\kappa) = 2I_{2j+1}(\kappa)/\kappa$ with the exact composition law $K_a * K_b = K_{a+b}$ — an identity the Wilson action does not share, which is why the uniformity result of Section VII requires the switch.

All numerical enclosures below use directed rational interval arithmetic (no floating point in any verdict); Bessel-function values are enclosed via their positive power series with a certified geometric tail bound, and transcendental combinations (logs, exponentials) via certified two-sided Taylor or continued-fraction enclosures with explicit remainder bounds. Every enclosure states its width, and every stage includes at least one deliberately falsified control (a wrong-recurrence Bessel tamper, a coupling substitution, a sign flip) that must, and does, produce a separated bracket.

IV. THE FREE REDUCED GAP

Theorem 1 (YM-1)**.** *For the normalized single-link Wilson convolution operator with $\beta = 2$, the reduced spectral gap*

$$\Delta_{\mathrm{red}}(a{=}1, \beta{=}2) = -\log\big(I_2(2)/I_1(2)\big) \qquad (2)$$

lies in the certified enclosure $[0.83672330623158891305006780105 4\dots]$, *width* $< 10^{-30}$*, an exact rational interval.*

Proof: $\lambda_j(\beta) = I_{2j+1}(\beta)/I_1(\beta)$ is exact by the character expansion of K_β; $\lambda_{1/2} \in (0, 1)$ strictly (control C1), so the gap is strictly positive; a tampered substitution $I_2 \to I_3$ separates the bracket (control C2); a width budget is enforced as a hard failure condition (control C3). This is the program's base case and the only place where a bare single-link spectrum, rather than a graph transfer operator, is the object of study.

V. THE INTERACTING THETA-GRAPH GAP AT FINITE COUPLING

The interacting transfer operator on B_3 is $T_\kappa = M^{1/2}(K \times K)M^{1/2}$, where K is the single-link kernel of Section III and $M = M_\kappa$ is the plaquette interaction weight $e^{(\kappa/2)\mathrm{Tr}U}$ on the third (closing) face.

Theorem 2 (YM-2: sandwich bound)**.** *For $0 \le \kappa < \kappa_0 = \Delta_{\mathrm{red}}/6$, the min-max sandwich $m_- T_0 \preceq T_\kappa \preceq m_+ T_0$ holds with $m_+/m_- = e^{6\kappa}$, using $|\mathrm{Tr}U| \le 2$. At $\kappa = 1/8$ this certifies a reduced gap $\ge 0.08672330623\dots$.*

Theorem 3 (YM-3: first-order crossing direction)**.** *The free top-excited eigenspace is exactly two-dimensional, $\{\chi_{1/2}(A), \chi_{1/2}(B)\}$, certified separated from $\chi^2_{1/2}, \chi_1, \chi_{3/2}$. The exact theta integral $\int \chi_{1/2}(A)\chi_{1/2}(B)\chi_{1/2}(AB^{-1}) = 1/2$ splits this doublet at first order into eigenvalues shifted by $\pm\lambda_{1/2}/4$; the vacuum first derivative is exactly* 0*. Hence $r'(0) = \lambda_{1/2}/4 \in [0.10828185668\dots]$, and the crossing direction is the single symmetric line $\chi_{1/2}(A) + \chi_{1/2}(B)$ — rank one, and invariant under the graph's $A \leftrightarrow B$ symmetry.*

Theorem 4 (YM-4: symmetry protection)**.** *Let $S : (A, B) \mapsto (B, A)$. Then $[S, T_\kappa] = 0$ for* every *κ, because $\mathrm{Tr}(BA^{-1}) = \mathrm{Tr}(AB^{-1})$ on* SU(2) *(real characters). Consequently the crossing line identified at first order in Theorem 3 cannot rotate out of the swap-even sector at any coupling: this is a theorem, not a first-order accident. Certified finite-κ spectral data, obtained from the exact expansion $e^{(\kappa/2)\mathrm{Tr}U} = \sum_j d_j f_j(\kappa)\chi_j$ with exact Clebsch–Gordan ring arithmetic and a certified truncation remainder, shows the swap-even branch rising and the swap-odd branch falling with κ on the sampled grid.*

Theorem 5 (YM-5: certified two-sided gap beyond the sandwich)**.** *Using the exact doubling identity $m_\kappa^2 = m_{2\kappa}$, a Weyl bound $\|PMQ\|^2 \le \lambda_{\max}(Pm_{2\kappa}P - (Pm_\kappa P)^2)$ and $\|QSQ\| \le \lambda^2_{1/2} e^{3\kappa}$ certify a two-sided gap at couplings beyond κ_0: at $\kappa = 1/8$, ratio ≤ 0.5003, gap ≥ 0.6924; at $\kappa = 1/4$ (past κ_0), ratio ≤ 0.5610, gap ≥ 0.5781; at $\kappa = 3/10$, gap ≥ 0.5332; the route fails closed at $\kappa = 1$.*

Theorem 5 is, structurally, an unconditional realization of the Schur-estimate-plus-positivity ("Kill Lemma") skeleton of an earlier, conjectural manuscript by the author that assumed the existence of a vacuum state; no conjecture, claim, or continuum statement from that manuscript is imported here (see the repository's `LINEAGE.md` for the explicit non-import ruling). Only the block-Schur-plus-positivity *shape* of the argument survived, now realized with certified exact-interval arithmetic on the graph carrier.

Theorem 6 (YM-6/YM-7: exact seam-integer counting)**.** *On the native five-dimensional carrier $V_5 = \{1, \chi_{1/2}(A), \chi_{1/2}(B), \chi_{1/2}(A)\chi_{1/2}(B), \chi_{1/2}(AB^{-1})\}$ (all exact eigenvectors of T_0, exact Gram matrix with a single* *off-diagonal overlap* $1/2$*), a Haynsworth congruence — the same finite-to-infinite compression device used elsewhere in this program for a Birman–Schwinger-type reduction — turns the gap statement into an* exact integer count $k_\Sigma(\kappa,\mu) = \#\{\mathrm{spec}(T_\kappa) > \mu\}$*, computed via two interval-LDL inertia evaluations of a rational pencil. Certified exact counts:* $k = 1$ *at* $(\kappa,\mu) = (1/8, 3/5), (1/4, 3/5), (1/2, 1)$*, i.e.* $\lambda_2 \le \mu < \lambda_1$ exactly *— a strictly stronger statement than a numerical bracket. Enlarging the carrier to* $V_7 = V_5 \cup \{\chi_1(A), \chi_1(B)\}$ *(Gram block-diagonal, all seven vectors exact* T_0*-eigenvectors) lowers the complement top eigenvalue from* λ_1 *to* $\lambda_1\lambda_{1/2}$ *and unlocks the previously undecidable cell: exact count* $k = 1$ *at* $(7/10, 13/10)$*. On* V_5 *the cell* $(7/10, 5/4)$ *is honestly* REFUSED*, bracket* $[1,5]$*: the carrier is too small to decide it, and this is reported as a refusal rather than resolved by widening tolerance.*

VI. CAPSTONE: A GAP AT EVERY COUPLING ON THE FIXED GRAPH

Theorem 7 (YM-8)**.** *The interacting theta-graph transfer operator has a strictly positive spectral gap at* every *coupling* $\kappa \ge 0$*, on the fixed graph* B_3 *at fixed lattice spacing.*

Proof: a certified pointwise kernel floor $k \ge e^{-3\kappa}\big[c_0^{-1}e^{-\beta}\big]^2 > 0$ (exact enclosures, positive-series Bessel bounds) combined with the classical Jentzsch/Krein-Rutman theorem [2, 3] — cited, not rederived, per the citation discipline stated in Section IX — gives a simple Perron eigenvalue, hence a strict gap, at every κ. The window $\kappa \in [0, 7/10]$ additionally carries the exact counts and enclosures of Theorems 2 –6.

Remark 1. *We state as plainly as possible what Theorem 7 does* not *say. A positive gap at every fixed coupling on a fixed, finite graph is compatible with the gap closing as the graph is refined or as the lattice spacing is taken to zero. Nothing here is uniform in the cutoff or the volume; that is exactly the content that Sections VII–VIII address on restricted families, and exactly the content that remains open in general (Section XI). A later result in this program (Theorem 9) further demotes the Jentzsch/Krein-Rutman anchor used here to a* simplicity-only *role, once positivity itself is re-derived from a manifest square; that correction is recorded in the governance ledger of the repository and does not alter Theorem 7's conclusion.*

VII. A FIRST UNIFORMITY-IN-THE-CUTOFF THEOREM

Every result so far is at one fixed lattice spacing a. We now obtain the program's first result that holds *for every* $a > 0$ simultaneously, by replacing the Wilson action with the heat-kernel action $K_a(g) = \sum_j d_j\, e^{-aC_j}\, \chi_j(g)$, $C_j = j(j+1)$ the Casimir. This action satisfies the exact semigroup law $K_a * K_b = K_{a+b}$: composing n links of spacing a/n reproduces one link of spacing a *exactly*, with no discretization error. The Wilson action has no such exact composition law, which is why it cannot support the argument below.

Theorem 8 (YM-9)**.** *The free reduced gap under the heat-kernel action is* $\Delta_{\mathrm{free}}(a) = C_{1/2} = 3/4$ *exactly, for every* $a > 0$ *— the transcendental* a*-dependence cancels identically. Along the declared linear trajectory* $\kappa(a) = \theta a$*, the interacting gap satisfies* $\Delta(a, \kappa(a)) \ge 3/4 - 6\theta$ *for every* $a > 0$*; at* $\theta = 1/16$ *this gives a cutoff-independent gap of exactly* $3/8$*, failing closed at* $\theta \ge 1/8$*. The seam count* $k(a, e^{-as})$ *is exactly* a*-independent (a combinatorial fact about which Casimir levels lie below a threshold), so the exact-counting machinery of Theorem 6 transports across the entire refinement family at once, not just at one spacing. By contrast, at fixed Wilson coupling* β *the reduced gap grows like* $1/a$ *and diverges as* $a \to 0$ *— there is no cutoff-independent gap on this action without a compensating running coupling.*

We state the honest remainder of Theorem 8 without softening. The trajectory $\kappa(a) = \theta a$ is *declared*, not derived: the physically motivated trajectory is fixed by asymptotic freedom, $\beta \sim \log(1/a)$, which is not modelled here. The graph is fixed, so this is a statement about refining the same graph's lattice spacing, not about infinite-volume growth. The heat-kernel action is itself a choice among a regulator class (Section IX addresses which parts of the program are regulator-independent). And Δ throughout is a reduced-graph transfer-operator gap, not a reconstructed physical mass gap in any continuum sense. What has moved is narrow but real: one structural gate — uniformity in the cutoff — goes from wholly untouched to touched, with an exact theorem, on a toy carrier.

VIII. VOLUME: WHERE UNIFORMITY EXTENDS AND WHERE IT PROVABLY FAILS

Theorem 9 (YM-12, volume half)**.** *Let* L_m *be the graph formed by vertex-gluing* m *copies of the theta graph with* no shared plaquettes*. The transfer operator tensor-factorizes exactly across the* m *copies, so the ratio* λ_2/λ_1 *is* m*-independent, and combined with Theorem 8,* $\Delta(L_m, a, a/16) \ge 3/8$ *for* every m *and* every a *— a gap uniform in volume and cutoff simultaneously, on this family.*

This closes the interacting half of the volume gate audited in Theorem 11 below, but only on the no-shared-face family. Theorem 11 shows the same sandwich-style route fails, exactly and quantifiably, once faces are shared; we record the two extremes as the boundaries of a genuine open problem, not as a resolved one.

The same paper (YM-12) also re-derives the positivity underlying Theorems 5–7 from a manifest square,

$T_\kappa(a) = S^*S$ with $S = [K_{a/2} \otimes K_{a/2}]\, m_{\kappa/2}$ — both factors being the program's own halved-parameter objects from Theorem 8's semigroup property and Theorem 5's doubling identity. This demotes the classical Jentzsch/Krein-Rutman citation used in Theorem 7 to a role of certifying only *simplicity* of the top eigenvalue, with positivity itself now square-sourced from within the program's own objects rather than imported.

IX. THREE STRUCTURAL GATES, AUDITED

We close the finite program with an audit of three structural questions on the toy-carrier family B_n (the "n-banana" graph: two vertices, n edges, $b_1 = n - 1$ holonomies, $F(n) = n(n-1)/2$ plaquettes; B_3 is the theta graph used throughout).

Theorem 10 (YM-11, Gate 1 — gauge: closed)**.** *Tree gauge-fixing on a connected graph is exact, with no Gribov obstruction: gauge-invariant states are exactly $L^2(G^{b_1})^{\mathrm{Ad}}$. For B_3 this certifies, rather than assumes, the carrier used in every capsule above.*

Theorem 11 (YM-11, Gate 2 — volume/infrared: split)**.** *The free reduced gap $C_{1/2} = 3/4$ is volume- and cutoff-uniform for every n and a: closed. The interacting sandwich bound gives $\Delta \geq C_{1/2} - 2F(n)\theta$, which degrades* quadratically *in n: at $\theta = 1/16$ the certified bounds are $3/8$ (n=3), exactly 0 (n=4), and $-1/2$ (n=5). The sandwich/envelope route is therefore proven* insufficient *for volume uniformity beyond Theorem 9's no-shared-face restriction, with the critical volume computed exactly rather than estimated.*

Theorem 12 (YM-11, Gate 3 — regulator universality: split)**.** *For the whole regulator class $K = \sum_j d_j c_j \chi_j$ with $c_j > 0$ (Wilson, heat-kernel, and others), the* counting *layer — content grading, the exact multiplicity law $m(j_1, j_2) = \min(2j_1, 2j_2) + 1$ on the Ad-invariant sector, carrier dimensions, and the source-restriction "blindness" ledger of Theorem 13 below — depends only on* $\mathrm{SU}(2)$ *representation theory and is therefore regulator-independent: closed. The* metric *layer, i.e. the numerical value of the gap, genuinely differs between regulators (witnessed directly by the Wilson vs. heat-kernel contrast in Theorem 8) and remains open.*

Theorem 13 (YM-10, source-restriction ledger)**.** *Every compressed carrier V used above is, in the sense of the companion source-restriction framework, a restriction of the full target space; the relevant question is not tightness of the resulting bound but whether V is* blind *to the target. The exact multiplicity law above independently reproduces $\dim V_5 = 5$ and $\dim V_7 = 7$ and re-derives the two-dimensionality of the $(\frac{1}{2}, \frac{1}{2})$ sector used in Theorem 3 from representation theory rather than from the Gram matrix. The exact probe count $b(V, s) = \mathrm{rank}(\Pi_s \mid \ker E)$ shows $b(V_5, s) = 0$ for $s \leq 2$ — the carrier used in Theorem 6 is exactly blindness-free within its own certified threshold window, which is why that choice of carrier was correct and not merely convenient. At $s = 3$, V_5 requires six probes and V_7 four; blindness is nonincreasing along $V_5 \subset V_7 \subset V_9$ with nonnegative stage increments; and the whole ledger, depending only on content and Casimir level, is exactly a-independent, so one ledger covers the entire refinement family of Theorem 8. This ledger is exact for the free target only; on interacting faces the content grading mixes, so $b(V, s)$ is a lower bound there, and closing that gap is an explicit named obligation of the program, not a silent one.*

We record one self-correction as part of the certification discipline itself: an earlier draft of Theorem 8's uniform seam count (T4) counted spectral *contents*, while the counting statement of Theorem 6 counts *eigenvalues with multiplicity*; at the relevant sector these differ (4 versus 5). Both quantities are independently a-independent, so the uniformity conclusion of Theorem 8 is unaffected, but the arithmetic was wrong and is corrected here. We regard catching and publishing this kind of error, rather than avoiding it, as part of what the certification discipline is for.

X. INTERACTION OVERLAP BETWEEN PLAQUETTES

Theorem 14 (YM-14)**.** *For a bridged pair of theta-graph blocks coupled by a single shared face, $e^{(\kappa/2)\mathrm{Tr}(A_1 A_2^{-1})}$ — the minimal case of genuine plaquette overlap not covered by Theorem 9's no-sharing family — the inter-block overlap seam is exactly rank-one and transported by the same graph symmetry and the same theta integral (1/2) that governs the intra-block seams of Theorem 3: one identity governs both. On the orthonormal carrier $W_5 = \{1, \chi_{1/2}(A_1), \chi_{1/2}(A_2), \chi_{1/2}(B_1), \chi_{1/2}(B_2)\}$, exact counts $k = 1$ are certified at $(\kappa, \mu) = (1/8, 3/5), (1/4, 7/10), (1/2, 9/10)$; at $\kappa = 1$ the carrier certifies count 0 at $\mu = 3/2$ (the compressed vacuum value is below μ), which we report as a carrier limitation rather than resolve by enlarging tolerance. The first-order overlap price is exactly $\kappa\lambda/4$, carried by a single swap-even line, in direct analogy with the single-block crossing direction of Theorem 3.*

This opens, but does not close, the object named by Theorem 11's Gate 2 as the real residual obstruction: Theorem 9 is uniform on no-shared-face volume growth, Theorem 11 shows the sandwich route fails on complete sharing, and a physically realistic bounded-degree lattice sits strictly between these two certified extremes. The next object named by the program is a one-dimensional block-transfer dock for a chain of bounded, but nonzero, overlap; it is not attempted here.

XI. WHAT REMAINS OPEN

We list the open content without euphemism, since the value of this program depends on the list being accurate.

- **Volume, general lattice.** Theorem 9 is uniform only on the no-shared-face family; Theorem 11 shows the sandwich route fails, with an exactly computed critical volume, once faces are shared; the bounded-overlap case of physical interest (Theorem 14) is opened but not closed.
- **Regulator universality, metric layer.** Only the combinatorial/counting layer of universality (Theorem 12) is closed; the numerical gap value differs by regulator and the asymptotically-free running-coupling trajectory needed to connect the heat-kernel result to the physical continuum limit is declared, not derived (Theorem 8).
- **Osterwalder–Schrader reconstruction.** No statement here constructs, or claims to construct, a continuum Wightman theory from these finite-graph transfer operators.
- **Non-triviality and the Clay predicate.** None of the above bears on the existence-and-mass-gap predicate of the Clay Millennium Problem as stated in [1]; that problem concerns the continuum, infinite-volume limit, which this program has not addressed and does not claim to have addressed.

XII. CONCLUSION

We have given, on the smallest graph carrying a Yang–Mills plaquette, a fully exact-arithmetic, machine-certified account of the interacting transfer operator's spectral gap: a strictly positive gap at every coupling on the fixed graph (Theorem 7), an exact eigenvalue-counting reformulation on part of the coupling range (Theorem 6), the program's first cutoff-uniform theorem (Theorem 8), and a volume-uniform extension on a restricted family with the extension's exact failure point identified on the complementary family (Theorems 9, 11). Each result is accompanied by negative controls that must, and do, fail when the claim is deliberately falsified, and each is reproducible from the certificates and scripts in the accompanying repository. The value we claim for this program is not proximity to the continuum problem, which remains entirely open, but a fully verified, error-caught, and error-corrected finite testbed — together with a precise catalogue, Section XI, of exactly which structural gates a genuine continuum proof would still need to close.

DATA AND CODE AVAILABILITY

All certificates, verification scripts, and test suites are available in the public repository `Parveen117/Publications`, directory `papers/yang-mills-certified-benchmark/`, under an open license. The theorem-to-script correspondence and reproduction instructions are given in full in Section II and Table I; no result in this paper depends on any data, computation, or software that is not in that public repository and independently re-executable by a reader.

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